

Aluminum involvement in the progression of Alzheimer's disease.

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The neuroanatomic specificity with which Alzheimer's disease (AD) progresses could provide clues to AD etiopathology. Magnetic resonance imaging studies of AD clinical progression have confirmed general conclusions from earlier studies of AD neuropathological progression wherein neurofibrillary tangle pathology was observed to spread along a well-defined sequence of corticocortical and corticosubcortical connections, preferentially affecting certain cell types, while sparing others. Identical and non-identical twin studies have consistently shown AD has mixed (environmental and genetic) etiopathogenesis. The decades-long prodromal phase over which AD develops suggests slow but progressive accumulation of a toxic or infective agent over time. Major environmental candidates are reviewed to assess which best fits the profile of an agent that slowly accrues in susceptible cell types of AD-vulnerable brain regions to toxic levels by old age, giving rise to AD neuropathology without rapid neuronal lysis. Chronic aluminum neurotoxicity best matches this profile. Many humans routinely ingest aluminum salts as additives contained in processed foods and alum-treated drinking water. The physical properties of aluminum and ferric iron ions are similar, allowing aluminum to use mechanisms evolved for iron to enter vulnerable neurons involved in AD progression, accumulate in those neurons, and cause neurofibrillary damage. The genetic component of AD etiopathogenesis apparently involves a susceptibility gene, yet to be identified, that increases aluminum absorption because AD and Down syndrome patients have higher than normal plasma, and brain, aluminum levels. This review describes evidence for aluminum involvement in AD neuropathology and the clinical progression of sporadic AD.